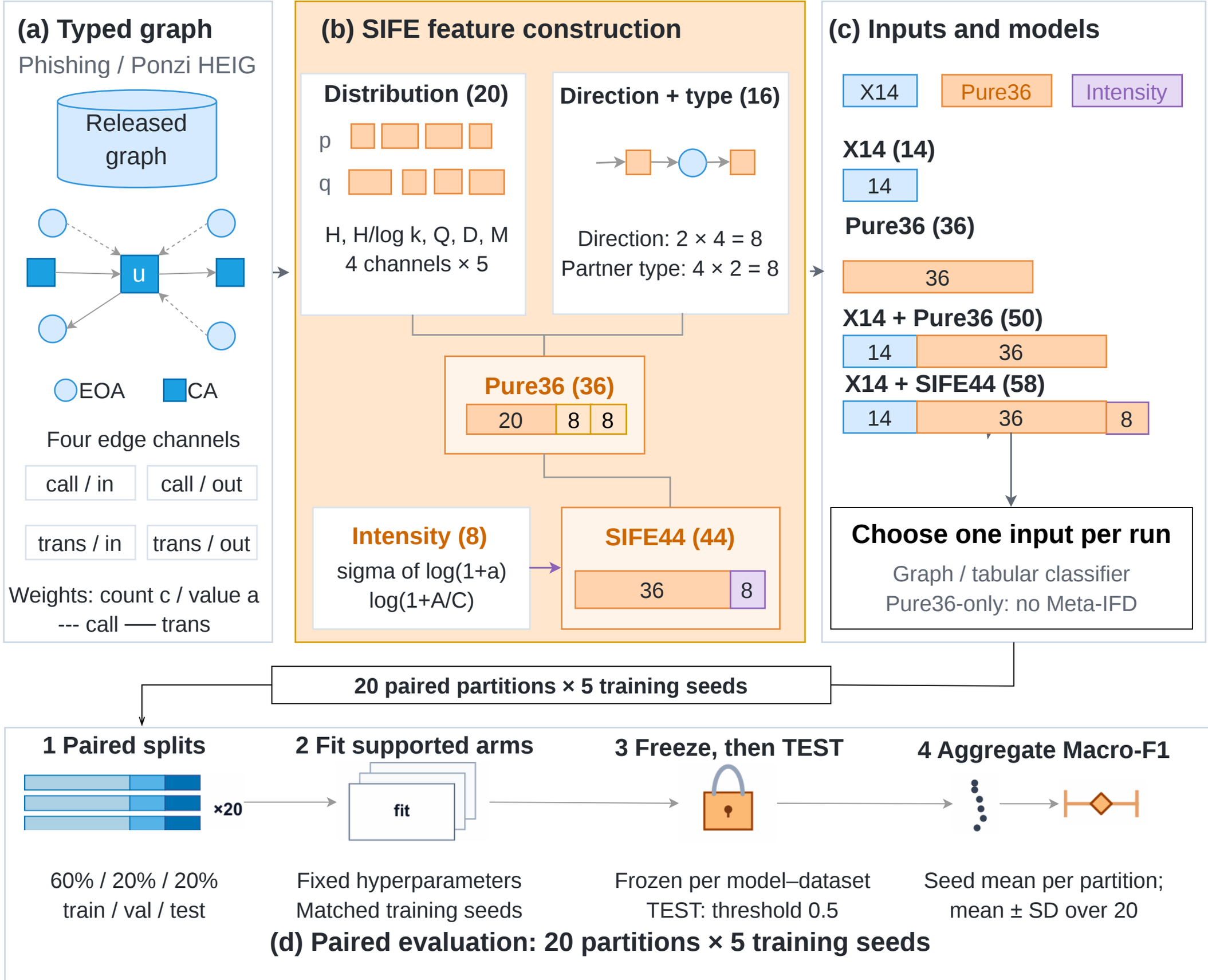

(a) Typed graph
Phishing / Ponzi HEIG
Released graph
u
EOA
CA
Four edge channels
call / in
call / out
trans / in
trans / out
Weights: count c / value a
--- call —— trans
(b) SIFE feature construction
Distribution (20)
p
q
H, H/log k, Q, D, M
4 channels × 5
Direction + type (16)
Direction: 2 × 4 = 8
Partner type: 4 × 2 = 8
Pure36 (36)
20
8
8
Intensity (8)
sigma of log(1+a)
log(1+A/C)
SIFE44 (44)
36
8
(c) Inputs and models
X14
Pure36
Intensity
X14 (14)
14
Pure36 (36)
36
X14 + Pure36 (50)
14
36
X14 + SIFE44 (58)
14
36
8
Choose one input per run
Graph / tabular classifier
Pure36-only: no Meta-IFD
20 paired partitions × 5 training seeds
1 Paired splits
×20
2 Fit supported arms
fit
3 Freeze, then TEST
4 Aggregate Macro-F1
60% / 20% / 20%
train / val / test
Fixed hyperparameters
Matched training seeds
Frozen per model–dataset
TEST: threshold 0.5
Seed mean per partition;
mean ± SD over 20
(d) Paired evaluation: 20 partitions × 5 training seeds